

REGIONAL PRODUCT INFORMATION (REGIONAL) ENGLISH

RHOPHYLAC, 300

HUMAN ANTI-D IMMUNOGLOBULIN

VERSION: 3.0

REVISION DATE: 29-Oct-2019

**REASON FOR CHANGE: CCDS update V2.0 & V3.0: Addition of dose
recommendation summary and editorial amendments**

Text Convention:

	Headers, footers and line numbers will not be printed.
Text:	text to be adapted as appropriate, according to local requirements

CSL Behring

1. NAME OF THE MEDICINAL PRODUCT

Rhophylac 300

2. QUALITATIVE AND QUANTITATIVE COMPOSITION

a. Active substance

Human Anti-D immunoglobulin*.

The product contains a maximum of 30 mg/ml of human plasma proteins of which 10 mg/ml is human albumin as stabiliser. At least 95% of the other plasma proteins are IgG. Distribution of the IgG subclasses (approximate values): IgG1 84.1 %, IgG2 7.6 %, IgG3 8.1 %, IgG4 1.0 %.

Rhophylac contains not more than 5 micrograms/ml IgA.

*Produced from the plasma of human donors.

b. Excipients

Human albumin	10 mg/ml
Glycine	20.6 mg/ml
Sodium chloride	≤ 250 mmol/l

Rhophylac contains no preservatives.

3. PHARMACEUTICAL FORM AND UNIT CONTENT OF ACTIVE INGREDIENT

Solution for intravenous or intramuscular injection in a pre-filled syringe.

Each 2 ml solution in a pre-filled syringe contains:

Human anti-D immunoglobulin G (IgG) 1,500 IU (300 micrograms) corresponding to a concentration of 750 IU (150 micrograms) per ml.

Rhophylac is a clear, colourless to pale yellow immunoglobulin solution. The solution is slightly hypertonic. Rhophylac has an osmolality of at least 240 mOsmol/kg.

4. INDICATIONS/POTENTIAL APPLICATIONS

Prevention of Rh(D) immunisation in Rh(D)-negative women

- *Antepartum prophylaxis:*

- Planned antepartum prophylaxis
- Antepartum prophylaxis following complications of pregnancy including :

Abortion/threatened abortion, ectopic pregnancy or hydatidiform mole, intrauterine fetal death, Transplacental haemorrhage resulting from antepartum haemorrhage, amniocentesis, chorionic biopsy or obstetric manipulative procedures e.g. external version, invasive interventions, cordocentesis, blunt abdominal trauma or fetal therapeutic intervention.

- *Postpartum prophylaxis:*

- Delivery of a Rh(D)-positive (D, Dweak, Dpartial) baby .

An Rh(D) incompatible pregnancy is assumed if the fetus/baby is either Rh(D) positive or Rh(D) unknown or if the father is either Rh(D) positive or Rh(D) unknown.

Treatment of Rh(D)-negative adults, children and adolescents (0-18 years) after incompatible transfusions of Rh(D)-positive blood or other products containing red blood cells eg: platelet concentrate.

5. POSOLOGY AND METHOD OF ADMINISTRATION

The following dose schedules are recommended based on the clinical studies performed with Rhophylac, however consideration must be given to professional guidelines for the use of anti-D IgG in the individual country of application.

The dose of anti-D immunoglobulin should be determined according to the level of exposure to Rh(D) positive red blood cells (RBCs) and based on the knowledge that 0.5 ml of packed Rh(D)

positive RBCs or 1 ml of Rh(D) positive blood is neutralised by approximately 10 micrograms (50 IU) of anti-D immunoglobulin.

Dosage:

Prevention of Rh(D) immunisation in Rh(D)-negative women

- *Antepartum prophylaxis:* The recommended dose is a single dose of 300 micrograms (1,500 IU) administered by intravenous or intramuscular injection.
 - *Planned antepartum prophylaxis:*

A single dose of 300 micrograms at 28 to 30 weeks of pregnancy. If the need for antepartum prophylaxis is identified in the period beyond 30 weeks of pregnancy, Rhophylac should not be withheld but administered as soon as possible.
 - *Antenatal prophylaxis following complications of pregnancy:*

300 micrograms should be administered by intravenous or intramuscular injection as soon as possible within 72 hours after the at-risk event. If more than 72 hours have elapsed, the product should not be withheld but administered as soon as possible. If necessary, administration of anti-D IgG should be repeated every 6 to 12 weeks until the time of delivery.
- *Postpartum prophylaxis:* 300 micrograms (1500 IU) should be administered as soon as possible within 72 hours of delivery of an Rh(D) positive (D, D^{weak}, D^{partial}) infant, by the intravenous or intramuscular route. If the 72-hour limit is exceeded, Rhophylac should be administered anyway. The postpartum dose must be given even when antepartum prophylaxis has been administered and even if residual activity from antepartum prophylaxis can be demonstrated in maternal serum.

If a large foeto-maternal haemorrhage (haemorrhage volume greater than 4 ml of Rh(D) positive foetal blood) is suspected, e.g., in the event of foetal anaemia or intrauterine foetal death, its extent should be determined by a suitable method, e.g. Kleihauer-Betke test, and additional doses of anti-D should be administered as indicated (10 micrograms (50 IU) for each 0.5 ml of Rh(D) positive foetal RBCs or per 1ml Rh(D) positive foetal blood).

Incompatible transfusions of Rh(D) positive RBCs in Rh(D) negative patients

The recommended dose is 10 micrograms (50 IU) anti-D immunoglobulin per 0.5 ml of Rh(D) positive erythrocyte concentrate or 1 ml of transfused Rh(D) positive blood. The appropriate dose should be determined in consultation with a specialist in blood transfusion. Follow-up tests for Rh(D) positive RBCs should be done every 48 hours and further anti-D administered until all Rh(D) positive RBCs have cleared from the circulation. The intravenous route of administration is recommended as it will achieve adequate plasma levels immediately. If given by intramuscular administration the large doses should be administered over a period of several days. A maximum dose of 3,000 micrograms (15000 IU) is sufficient in the case of larger incompatible transfusions independent of whether the transfusion volume is greater than 300 ml of Rh(D)-positive blood or 150 ml of Rh(D) positive erythrocyte concentrate was transfused. Due to possible risk of haemolysis, however, it is suggested to not exceed the dose of 3000 micrograms (15000 IU).

The dose recommendations for prevention of Rh(D) isoimmunisation are summarized in the following table:

Indication	Timing of Administration	Dose
<u>Antepartum prophylaxis:</u>		
Planned antepartum prophylaxis	At 28 to 30 weeks of pregnancy	300 micrograms (1500 IU) as single dose
Antepartum prophylaxis following complications of pregnancy	Within 72 h of complication	300 micrograms (1500 IU) as single dose [†]
<u>Postpartum prophylaxis:</u>		
	Within 72 h of birth	300 micrograms (1500 IU) as single dose [†]
Large feto-maternal haemorrhage (> 4 ml)	Within 72 h of complication	300 micrograms (1500 IU) as single dose plus: • 10 micrograms (50 IU) per 0.5 ml Rh(D) positive foetal RBCs or • 10 micrograms (50 IU) per 1 ml Rh(D) positive foetal blood

Incompatible transfusions	Within 72 h of exposure	• 10 micrograms (50 IU) per 0.5 ml transfused Rh(D) positive RBCs concentrate or • 10 micrograms (50 IU) per 1 ml transfused Rh(D) positive blood
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†The Rophylac dose may need to be increased if the patient is exposed to > 15 ml of Rh(D) positive foetal RBCs. In this case, follow the dosing guidelines for large fetomaternal haemorrhage.

Paediatric population

As the posology in case of incompatible transfusion depends on the volume of Rh(D) positive blood or Rh(D) positive RBC concentrate transfused, the recommended dose in children and adolescents (0-18 years) is not considered to be different to that of adults. However, the appropriate dose should be determined in consultation with a specialist in blood transfusion.

Elderly population

As the posology in case of incompatible transfusion depends on the volume of Rh(D) positive blood or Rh(D) positive RBC concentrate transfused, the recommended dose in elderly patients (≥65 years of age) is not considered to be different to that of adults. The appropriate dose, however, should be determined in consultation with a specialist in blood transfusion.

Method of administration:

For intravenous or intramuscular use.

As with all blood products, patients should be observed for at least 20 minutes following administration of Rophylac.

Rhophylac can be administered by slow intravenous or intramuscular injection.

In case of haemorrhagic disorders where intramuscular injections are contraindicated, Rhophylac should be administered intravenously. If large doses (>5 ml) are required and intramuscular injection is chosen, it is advisable to administer them in divided doses at different sites.

Obesity

In patients with a body mass index (BMI) ≥ 30 intravenous administration should be considered (see section “Warnings and precautions”).

6. CONTRAINDICATIONS

Hypersensitivity to the active substance or to any of the excipients of Rhophylac (see section “Qualitative and quantitative composition”).

Hypersensitivity to human immunoglobulins.

The intramuscular route is contraindicated in persons with severe thrombocytopenia or other disorders of haemostasis.

7. WARNINGS AND PRECAUTIONS

In the case of postpartum use, anti-D immunoglobulin is intended for maternal administration. It should not be given to the newborn infant.

The product is neither intended for use in Rh(D)-positive individuals nor for individuals already immunised to Rh(D) antigen.

Hypersensitivity / Anaphylaxis

Allergic responses to anti-D immunoglobulin may occur even in patients who have tolerated previous administrations. Patients should be informed of the early signs of hypersensitivity reactions including hives, generalised urticarial, tightness of the chest, wheezing, hypotension and anaphylaxis. The treatment required depends on the nature and severity of the side effect. In case of shock, the current medical standards for treatment of shock should be observed. If symptoms of allergic or anaphylactic type reactions occur, immediate discontinuation of the administration is required.

The concentration of IgA in Rophylac was found to be below the detection limit of 5 micrograms/ml. Nevertheless, the product may contain trace amounts of IgA. Although anti-D immunoglobulin has been used successfully to treat selected IgA deficient patients, individuals who are deficient in IgA have the potential for developing IgA antibodies and may have anaphylactic reactions after administration of blood components containing IgA. The physician must therefore weigh the benefit of treatment with Rophylac against the potential risks of hypersensitivity reactions.

Haemolytic reactions

Patients in receipt of an incompatible transfusion who receive very large doses of anti-D immunoglobulin should be monitored clinically and by biological parameters because of the risk of haemolytic reaction.

Obesity

There have been reports that the intramuscular administration of Rophylac in patients with a body mass index (BMI) ≥ 30 is associated with a risk of lack of efficacy. Therefore, in patients with a BMI ≥ 30 intravenous administration should be considered.

Pathogen safety

Standard measures to prevent infections resulting from the use of medicinal products prepared from human blood or plasma include selection of donors, screening of individual donations and plasma pools for specific markers of infection and the inclusion of effective manufacturing steps for the inactivation/removal of viruses. Despite this, when medicinal products prepared from human blood or plasma are administered, the possibility of transmitting infective agents cannot be totally excluded. This also applies to unknown or emerging viruses and other pathogens.

The measures taken are considered effective for enveloped viruses such as human immunodeficiency virus (HIV), hepatitis B virus (HBV) and hepatitis C virus (HCV). They may be of limited value against non-enveloped viruses such as hepatitis A (HAV) or parvovirus B19.

There is reassuring clinical experience regarding the lack of hepatitis A or parvovirus B19 transmission with immunoglobulins and it is also assumed that the antibody content makes an important contribution to the viral safety.

It is strongly recommended that every time that Rhophylac is administered to a patient, the name and batch number of the product are recorded in order to maintain a link between the patient and the batch of the product.

8. INTERACTION

Live attenuated virus vaccines

Active immunisation with live virus vaccines (e.g. measles, mumps, rubella or varicella) should be postponed until 3 months after the last administration of anti-D immunoglobulin, as the efficacy of the live virus vaccine may be impaired. If anti-D immunoglobulin needs to be administered within 2 to 4 weeks of a live virus vaccination, then the efficacy of such a vaccination may be impaired.

Interference with serological testing

After injection of immunoglobulin the transitory rise of the various passively transferred antibodies in the patient's blood may result in misleading positive results in serological testing. Passive transmission of antibodies to erythrocyte antigens, e.g. A, B, Rh(C), Rh(D) may interfere with some serological tests for RBC antibodies e.g. the antiglobulin test (Coombs' test), particularly in Rh(D) positive neonates whose mothers have received antepartum prophylaxis.

9. PREGNANCY AND LACTATION

This medicinal product is used in pregnancy and postpartum.

This medicinal product can be used during breastfeeding.

10. EFFECTS ON ABILITY TO DRIVE AND USE MACHINES

Rhophylac has no influence on the ability to drive and use machines.

11. UNDESIRABLE EFFECTS

Summary of the safety profile

When anti-D immunoglobulins are administered by the intramuscular route, local pain and tenderness can be observed at the injection site.

Tabulated list of adverse reactions

The following adverse reactions have been reported from 592 patients in clinical studies and from post-marketing experience. The summary table presented below is according to the MedDRA system organ classification (SOC and Preferred Term Level).

Frequency has been evaluated using the following criteria: Very common ($\geq 1/10$), Common ($\geq 1/100$ and $< 1/10$), Uncommon ($\geq 1/1,000$ and $< 1/100$), Rare ($\geq 1/10,000$ and $< 1/1,000$), Very rare ($< 1/10,000$).

System Organ Class	Adverse Reaction	Frequency of the Adverse Reaction
Immune system disorders	Hypersensitivity reactions, Anaphylactic shock	Rare
Nervous system disorders	Headache	Uncommon
Cardiac disorders	Tachycardia	Rare
Vascular disorders	Hypotension	Rare
Respiratory, thoracic and mediastinal disorders	Dyspnoea	Rare
Gastrointestinal disorders	Nausea, Vomiting	Rare
Skin and subcutaneous tissue	Skin reaction, Erythema,	Uncommon

System Organ Class	Adverse Reaction	Frequency of the Adverse Reaction
disorders	Pruritus	
Musculoskeletal and connective tissue disorders	Arthralgia	Rare
General disorders and administration site conditions	Pyrexia, Malaise, Chills	Uncommon
	At injection site: Swelling, Pain, Erythema, Induration, Warmth, Pruritus, Rash	Rare

There have been spontaneous reports of severe intravascular haemolysis when anti-D has been administered intravenously to Rh(D)-positive ITP patients. Haemolysis resulting in death has been reported.

The exact frequency of this adverse event is not known.

For safety information with respect to transmissible agents, see section “Warnings and precautions”.

Reporting of suspected adverse reactions

Reporting suspected adverse reactions after authorisation of the medicinal product is important. It allows continued monitoring of the benefit/risk balance of the medicinal product. Healthcare professionals are asked to report any suspected adverse reactions.

12. OVERDOSE

No data are available on overdosage. Consequences of an overdose are not known.

13. PROPERTIES / EFFECTS

ATC Code: J06BB01.

Pharmacotherapeutic group: immune sera and immunoglobulins: Anti-D (Rh) immunoglobulin.

Mechanism of action / Pharmacodynamics

Rhophylac contains specific IgG antibodies against the Rh(D) antigen of human erythrocytes.

Prevention of Rh(D) isoimmunisation During pregnancy, and especially at the time of childbirth, foetal red blood cells may enter the maternal circulation. When the woman is Rh(D)-negative and the foetus Rh(D)-positive, the women might become immunised to the Rh(D) antigen and may produce anti-Rh(D) antibodies which cross the placenta and may cause haemolytic disease of the newborn. Passive immunisation with anti-D immunoglobulin prevents Rh(D) immunisation in more than 99% of cases provided that a sufficient dose of anti-D immunoglobulin is administered early enough after exposure to Rh(D)-positive foetal red blood cells.

The mechanism by which anti-D immunoglobulin suppresses immunisation to Rh(D)-positive RBCs is not known. Suppression may be related to the clearance of the Rh(D) positive RBCs from the circulation before they reach immunocompetent sites or, it may be due to more complex mechanisms involving recognition of foreign antigen and antigen presentation by the appropriate cells at the appropriate sites in the presence or absence of antibody.

In clinical studies in Rh(D)-negative healthy male volunteers, both the intravenous and intramuscular administration of Rhophylac resulted in an efficient clearance of Rh(D)-positive erythrocytes from the circulation. While the intravenous administration of Rhophylac caused an instant onset of red blood cell disappearance, the onset of elimination of red blood cells following intramuscular administration was delayed as anti-D IgG had to be first absorbed from the injection site. On an average, 70% of injected RBCs were cleared 2 hours after intravenous administration of Rhophylac. After intramuscular administration, a similar degree of RBC clearance was measured after 12 hours.

Clinical efficacy

The efficacy, safety and pharmacokinetics of Rophylac are supported by the results of 3 clinical studies that have shown that ante- and postpartum administration of Rophylac effectively prevents Rh(D) immunization. In one clinical study, Rophylac 200 micrograms (1000 IU) was administered postpartum in 139 per protocol subjects. In the other 2 clinical studies, Rophylac 300 micrograms (1500 IU) was administered antepartum in 408 per protocol subjects and in addition postpartum (6 to 11 months after the birth) in 256 subjects who gave birth to an Rh(D)-positive baby. None of the pregnant women included in these studies developed antibodies against the Rh(D) antigen. In two clinical studies in Rh(D)-negative women, Rophylac was administered in 28th week of pregnancy and within 72 hours after the birth of a Rh(D)-positive child. 207 per protocol subjects were given the antepartum dose of Rophylac 300 intravenously, and 201 per protocol subjects were given it intramuscularly. In more than 99% of cases, the method of post- and antepartum administration was the same.

Paediatric population

The safety and efficacy of Rophylac have not been established in clinical studies in paediatric patients after incompatible transfusion of Rh(D) positive blood or other products containing Rh(D) positive RBCs.

14. PHARMACOKINETICS

Absorption and Distribution

Measurable levels of antibodies are obtained approximately 4 hours after intramuscular injection. Human anti-D immunoglobulin for intramuscular administration is slowly absorbed into the recipients's circulation and reaches a maximum after a delay of 2 to 3 days. Measurable levels of antibodies are obtained immediately after intravenous injection. IgG is quickly distributed between plasma and extravascular fluid. Two to 3 weeks post injection, serum levels are aligned and a difference between the two routes of administration can no longer be detected.

Elimination

Human anti-D immunoglobulin has a half-life of about 3 to 4 weeks. This half-life may vary individually from patient to patient. The mean half-life in the circulation of pregnant women with normal IgG levels was 17 days.

IgG and IgG-complexes are broken down in cells of the reticuloendothelial system.

15. PRECLINICAL DATA

Due to induction of and interference with antibodies, there are limited preclinical data of relevance for anti-D immunoglobulin.

Repeated dose testing and embryo-foetal toxicity studies have not been conducted and are impracticable. The potential for mutagenic effects of immunoglobulins have not been studied.

16. OTHER INFORMATION

Incompatibilities

In the absence of compatibility studies, Rhophylac must not be mixed with other medicinal products.

Shelf life

3 years

Special precautions for storage

Store in a refrigerator (+2 to +8 °C). Do not freeze.

Keep the syringe (originally blistered) in the outer carton in order to protect from light.

Store out of the reach and sight of children.

Do not use after the expiry date stated on both the syringe label and the outer carton.

Nature and contents of container

Glass syringe (type I glass acc. To Ph. Eur.) pre-filled with 2 ml solution for injection (1,500 IU anti-D IgG).

Instructions for use and handling and disposal

Rhophylac should be brought to room temperature (25 °C) before use.

Rhophylac should be inspected visually for particulate matter and discolouration prior to administration. The solution should be clear or slightly opalescent. Do not use solutions which are cloudy or have deposits.

Use only once (one syringe – one patient).

Any unused product or waste material should be disposed of in accordance with local requirements.

17. PACKS

2 ml (300 micrograms) solution in a pre-filled syringe with 1 injection needle

18. DATE OF REVISION OF THE TEXT

June 2021

19. MANUFACTURER

CSL Behring AG

Bern, Switzerland

Note: Rhophylac® is a registered trademark of CSL Behring AG in many countries.

HISTORY OF REVISIONS

Version No.	Revision Date	Reason for Change
n.a.	04-JUL-2007 /	New document according to CCDS Rev. 04-JUL-2007
n.a.	09-NOV-2010 /	Addition of trademark clause
n.a.	October-2014:	Addition of a warning for obese patients in case of IM use and update to the latest Labeling Changes in Switzerland (CoO)
1.0	05/10/2015	CCSI V2.0 Update and new RPI template implementation.
2.0	19 Apr 2017	Correction to CCSI V2.0 Update and new RPI template implementation.
3.0	29 Oct 2019	CCDS update V2.0 & V3.0: Addition of dose recommendation summary and editorial amendments